[METRIC]

A-A-59315/21
September 4, 1998
SUPERSEDING
MIL-P-85856/21
December 7, 1988

COMMERCIAL ITEM DESCRIPTION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL-CHROMIUM-ALUMINUM, TYPE I, CLASS 2

The General Services Administration has authorized the use of this commercial item description sheet as a replacement for MIL-P-85856/21 by all federal agencies.

The complete requirements for procuring the powder material described herein shall consist of this document and the issue in effect of commercial item description A-A-59315/GEN.

1. SCOPE. This commercial item description sheet (CID) covers a nickel-chromium-aluminum powder that is applied using either plasma arc or combustion gas (flame spray) techniques.

2. SALIENT CHARACTERISTICS.

2.1 Chemical composition. Chemical composition shall conform to the following percentages, by weight, as shown in table I.

2.2 Particle size distribution. The particle size distribution shall conform to table II.

3. REGULATORY REQUIREMENTS. Regulatory requirements shall be in accordance with A-A-59315/GEN.

4. QUALITY ASSURANCE PROVISIONS. Quality assurance provisions shall be in accordance with A-A-59315/GEN.

5. PACKAGING. Packaging shall be in accordance with A-A-59315/GEN.

6. NOTES. The notes specified in A-A-59315/GEN are applicable to this CID sheet.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100.

FSC 3439

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TABLE I. Chemical composition. 1/, 2/

Element	Percentage	
	min.	max.
Chromium	15.5	21.5
Nickel	69.5	--
Iron	--	1.5
Aluminum	4.5	7.5
Carbon	--	0.3
Manganese	--	3.0
Silicon	--	2.0

1/ Percentages by weight are after removal of binder. Powder shall be a composite of not less than 96.5 percent by weight of an aggregate of nickel-chromium and aluminum powders with not more than 3.5 percent by weight of phenolic binder material capable of volatilizing completely at 538°C (1000°F).

2/ Powder particles shall consist of a composite structure of nickel-chromium alloy core surrounded by aluminum.

TABLE II. Particle size distribution by weight (percent).

Sieve Designation		Weight passed through sieve	Weight retained on sieve
Mesh	Microns (μm)		
100	150	--	1.0 max.
120	125	--	10.0 max.
325	45	20.0 max.	--

MILITARY INTERESTS:

Custodians:
Army - MR
Navy - AS
Air Force - 11
DLA - GS

Review activities:
Navy - SH
Air Force - 71, 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA/FSS-7FXE

Preparing activity:
Navy - AS
(Project 3439-0931)

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